

PRACTICE WORKBOOK

Leverage on Both Sides of the Market

A Workbook on Margin Trading, Margin Calls, and Short Selling

Numeric Problems and Applications for Investment Decision-Making

Prepared for MBA, M.Com., and MSc Finance Programs
Securities Analysis and Portfolio Management

Covers: Buying on Margin · Margin Calls · Short Selling · Leverage · Sensitivity Analysis

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Part I: Conceptual and Formula Foundation

1.1 Leverage, Equity, and the Architecture of a Margin Position

When an investor buys securities on margin, personal equity capital is combined with borrowed funds from a broker to acquire a larger position than equity alone would permit. Every percentage movement in the share price is applied to the full position, yet that gain or loss must be measured against the smaller equity base the investor contributed. This mismatch between the size of the exposure and the size of the capital base is the defining mechanism of leverage, and it is the reason margin trading amplifies both returns and losses relative to an unleveraged position.

Equity in a margin account is simply the market value of the position net of the broker's loan. Because the loan is fixed in dollar terms while the market value fluctuates with price, equity — and the percentage margin it represents — erodes automatically whenever the price declines. The initial margin is the percentage margin the investor holds when the position is opened; the maintenance margin is the minimum percentage margin the broker will accept thereafter. A margin call arises the instant the percentage margin falls to the maintenance level, at which point the broker demands additional cash or securities, or will liquidate part of the position to restore compliance.

1.2 The Mechanics and Obligations of a Short Sale

A short sale reverses the usual order of a securities transaction: the investor borrows shares from a broker or another investor's holdings, sells them immediately, receives cash proceeds, and only later repurchases equivalent shares to return them and close out the loan. Between the sale and the eventual repurchase, the short seller owes shares, not a cash liability whose dollar value moves directly with the stock's market price.

Because exchange rules prevent short sellers from withdrawing or investing the sale proceeds, those proceeds are held by the broker and combined with a separate margin deposit to satisfy the initial margin requirement on the trade. The short seller also carries a continuing obligation to compensate the share lender for any dividends distributed while the position remains open, since the lender would otherwise have received that income directly. These two features — restricted proceeds and dividend pass-through — have no counterpart in an ordinary long purchase and must be built into every short-sale calculation.

1.3 Asymmetric Risk Between Long-Margin and Short Positions

A long-margin position has a bounded downside: the maximum possible loss is the total capital committed, own funds plus borrowed funds, because a share price cannot fall below zero. Its upside, by contrast, is theoretically unbounded, since there is no ceiling on how high a share price can rise. A short position exhibits the mirror image of this profile. The maximum possible gain is capped and achieved only if the price falls all the way to zero, whereas the maximum possible loss is theoretically unbounded because no ceiling limits how high a share price can rise while the position remains open. This asymmetry is the single most important risk-management fact a short-seller must internalize, and it explains why protective devices such as stop-buy orders are used far more routinely on the short side of the market than on the long side.

1.4 The Role of Sensitivity Analysis in Margin Decision-Making

Because ROI, account equity, and margin adequacy are highly sensitive to the path the share price actually takes, a single-point estimate based on a single assumed ending price captures only part of the risk picture. Systematic sensitivity analysis — recalculating ROI and margin ratios across an entire range of hypothetical prices — allows an analyst to locate break-

even levels and margin-call thresholds before they are reached, to quantify precisely how far an adverse price move can travel before risk-control action becomes necessary, and to support decisions on position sizing, initial margin selection, and hedge design. The following formulas build toward exactly this capability, culminating in Group 6, which generalizes every prior relationship across an assumed range of prices.

1.5 Classification of Calculations

The thirty-two relationships used throughout this workbook fall into six logically connected groups. Group 1 establishes the size and financing structure of a position — how much stock is controlled and how levered it is. Group 2 measures the cash flows a position generates: capital gains, dividends, and financing costs, culminating in net income. Group 3 converts net income into a percentage return on the investor's own capital and contrasts leveraged with unleveraged outcomes. Group 4 values the position at any point in time and calculates the equity and margin ratio that determines collateral adequacy. Group 5 identifies the exact critical stock price—the margin-call threshold—at which the maintenance requirement is breached, along with the resulting decision rule. Group 6 extends every prior relationship across a range of hypothetical prices, enabling sensitivity analysis that anticipates risk before it materializes rather than reacting after the fact.

Group 1: Position Construction and Financial Exposure

This group establishes the size, financing, and exposure of long-margin and short-sale positions.

1. Total Stock Purchased:

$$\text{Total Stock Purchased} = \text{Initial Equity Investment} + \text{Amount Borrowed}$$

Defines the full dollar size of a margin-financed long position, combining the investor's own funds with the broker's loan.

2. Shares Purchased:

$$\begin{aligned} \text{Shares Purchased} &= \\ &\frac{\text{Total Stock Purchased}}{\text{Initial Stock Price}} \end{aligned}$$

Converts the dollar position into the number of shares controlled, which drives every subsequent gain, loss, and dividend calculation.

3. Short-Sale Proceeds:

$$\text{Short-Sale Proceeds} = \text{Initial Stock Price} \times \text{Number of Shares Sold Short}$$

Represents the cash credited to the investor's account when borrowed shares are sold, forming the base against which margin adequacy is later measured.

4. Current Value of Shares Owed:

$$\text{Current Value of Shares Owed} = \text{Ending Stock Price} \times \text{Number of Shares Sold Short}$$

Captures the investor's repurchase liability at any point in time; because it moves with price rather than remaining fixed, it is the source of the short-seller's theoretically unlimited loss exposure.

5. Leverage Multiple:

$$\begin{aligned} \text{Leverage Multiple} &= \\ &\frac{\text{Total Stock Purchased}}{\text{Initial Equity Investment}} \end{aligned}$$

Initial Equity Investment

Measures how many dollars of market exposure each dollar of investor equity controls, directly indicating how strongly returns will be amplified in either direction.

6. Effective Initial Margin Percentage (Short Sale):

$$\text{Effective Initial Margin Percentage (Short Sale)} = \frac{\text{Initial Margin Deposit}}{\text{Short-Sale Proceeds}}$$

Verifies the actual collateral cushion the investor has posted relative to the broker's stated initial-margin requirement on the short sale.

Group 2: Income, Costs, and Profit Measurement

This group measures gains, losses, dividend effects, financing costs, and net income for both sides of the market.

7. Capital Gain or Loss on a Long Position:

$$\text{Capital Gain or Loss} = \text{Shares Purchased} \times (\text{Ending Stock Price} - \text{Initial Stock Price})$$

Isolates the pure price-driven component of profit or loss, before dividends or financing costs are considered.

8. Dividends Received:

$$\text{Dividends Received} = \text{Shares Purchased} \times \text{Cash Dividend per Share}$$

Adds the income component that a long position earns during the holding period, which partly offsets financing costs.

9. Interest on Margin Loan:

$$\text{Interest on Margin Loan} = \text{Amount Borrowed} \times \text{Annual Interest Rate} \times (\text{Holding Period in Months} \div 12)$$

Prorates the annual borrowing cost to the actual holding period, recognizing that leverage is never free.

10. Net Income from a Long-Margin Position:

$$\text{Net Income} = \text{Capital Gain or Loss} + \text{Dividends Received} - \text{Interest on Margin Loan}$$

Consolidates all cash flows into a single bottom-line profit or loss figure attributable to the investor.

11. Capital Gain or Loss on a Short Sale:

$$\text{Capital Gain or Loss} = \text{Number of Shares Sold Short} \times (\text{Initial Stock Price} - \text{Ending Stock Price})$$

Reverses the long-position logic, since the short-seller profits when the price falls and loses when the price rises.

12. Dividends Paid by the Short-Seller:

$$\text{Dividends Paid} = \text{Number of Shares Sold Short} \times \text{Cash Dividend per Share}$$

Reflects the short-seller's contractual obligation to compensate the share lender for dividends foregone during the loan.

13. Net Income from Short Sale:

$$\text{Net Income} = \text{Capital Gain or Loss} - \text{Dividends Paid}$$

Provides the short-sale equivalent bottom line, in which dividend obligations always reduce, never enhance, the outcome.

Group 3: Return and Leverage Analysis

This group evaluates the return earned on the investor's own capital and demonstrates how leverage changes the return profile.

14. ROI on a Long-Margin Position:

$$\text{ROI on a Long-Margin Position} = \frac{\text{Net Income}}{\text{Initial Equity Investment}}$$

Expresses profit or loss as a percentage of the capital the investor personally committed — the appropriate basis for judging a leveraged trade.

15. ROI on a Long-Margin Position(alternate formula):

$$\text{ROI on a Long-Margin Position} = \frac{\text{Shares Purchased} \times (\text{Ending Price} - \text{Initial Price}) + \text{Dividends Received} - \text{Interest on Margin Loan}}{\text{Initial Equity Investment}}$$

Unpacks the same ratio into its underlying drivers, enabling isolation of which component — price change, income, or financing cost — dominates the outcome.

16. Shares Purchased without Margin:

$$\text{Shares Purchased without Margin} = \frac{\text{Initial Equity Investment}}{\text{Initial Stock Price}}$$

Establishes the counterfactual, unleveraged position size using only the investor's own capital.

17. ROI without Margin:

$$\text{ROI without Margin} = \frac{\text{Shares without Margin} \times (\text{Ending Price} - \text{Initial Price}) + \text{Shares without Margin} \times \text{Dividend per Share}}{\text{Initial Equity Investment}}$$

Provides the benchmark return the investor would have earned without borrowing, against which the leveraged outcome is compared.

18. ROI on a Short Sale:

$$\text{ROI on a Short Sale} = \frac{\text{Net Income}}{\text{Initial Margin Deposit}}$$

Measures short-sale profitability relative to the collateral actually posted, not the full value of shares borrowed.

19. Expanded Short-Sale ROI:

$$\text{Expanded Short-Sale ROI} = \frac{\text{Shares Sold Short} \times (\text{Initial Price} - \text{Ending Price}) - \text{Dividends Paid}}{\text{Initial Margin Deposit}}$$

Decomposes short-sale ROI to show how price decline and dividend obligations jointly determine the percentage return on deposited margin.

Group 4: Account Equity and Margin Adequacy

This group determines the investor's remaining equity and the adequacy of collateral at the prevailing market price.

20. Ending Market Value of a Long Position:

$$\text{Ending Market Value} = \text{Shares Purchased} \times \text{Ending Stock Price}$$

Values the position at the close of the period, forming the basis for equity and margin-ratio calculations.

21. Equity in a Long-Margin Account:

$$\text{Account Equity} = \text{Ending Market Value} - \text{Amount Borrowed}$$

Represents the investor's true residual claim on the position once the fixed broker loan is repaid.

22. Current Margin Ratio for a Long Position:

$$\text{Current Margin Ratio for a Long Position} = \frac{\text{Ending Market Value} - \text{Amount Borrowed}}{\text{Ending Market Value}}$$

Tracks, at any price, the proportion of the position financed by the investor's own equity — the figure a broker compares against the maintenance requirement.

23. Equity in a Short-Sale Account:

$$\text{Short-Sale Account Equity} = \text{Initial Margin Deposit} + \text{Short-Sale Proceeds} - \text{Dividends Paid} - \text{Current Value of Shares Owed}$$

Combines all cash inflows and the mark-to-market liability to determine the investor's net claim in a short-sale account.

24. Current Short-Sale Margin Ratio:

$$\text{Current Short-Sale Margin Ratio} = \frac{\text{Short-Sale Account Equity}}{\text{Current Value of Shares Owed}}$$

Provides the short-sale analogue of the long-position margin ratio, indicating collateral adequacy relative to the shares still owed.

Group 5: Margin-Call Thresholds and Risk Control

This group identifies the critical stock price at which the maintenance-margin requirement is breached.

25. Margin-Call Price for a Long Position:

$$\text{Margin-Call Price for a Long Position} = \frac{\text{Amount Borrowed}}{\text{Shares Purchased} \times (1 - \text{Maintenance Margin Percentage})}$$

Solves directly for the critical stock price at which a long position's margin ratio equals the maintenance requirement.

26. Long-Position Margin-Call Decision Rule:

$$\begin{aligned} &\text{Long-Position Margin-Call Decision Rule:} \\ &\text{Margin Call, if Ending Price} \leq \text{Long Margin-Call Price} \\ &\text{Safe, if Ending Price} > \text{Long Margin-Call Price} \end{aligned}$$

Converts the numerical threshold into an operational decision rule: a falling price is the source of risk for a long position.

27. Margin-Call Price for a Short Sale:

$$\text{Margin-Call Price for a Short Sale} = \frac{\text{Initial Margin Deposit} + \text{Short-Sale Proceeds} - \text{Dividends Paid}}{\text{Shares Sold Short} \times (1 + \text{Maintenance Margin Percentage})}$$

Derives the price at which a short sale's margin ratio falls to the maintenance requirement; risk here arises from price increases, not decreases.

28. Short-Sale Margin-Call Decision Rule:

$$\begin{aligned} &\text{Short-Sale Margin-Call Decision Rule:} \\ &\text{Margin Call, if Ending Price} \geq \text{Short-Sale Margin-Call Price} \\ &\text{Safe, if Ending Price} < \text{Short-Sale Margin-Call Price} \end{aligned}$$

The inequality direction is deliberately opposite to the long-position rule, reflecting the mirrored risk exposure of a short position.

Group 6: Sensitivity and Decision Analysis

This group examines how changes in stock prices affect ROI, account equity, and margin adequacy.

29. Long ROI at an Assumed Stock Price:

$$\begin{aligned} &\text{Long ROI at an Assumed Stock Price} = \\ &\frac{(\text{Shares Purchased} \times \text{Assumed Stock Price}) - \text{Total Stock Purchased} + \text{Dividends Received} - \text{Interest on Margin Loan}}{\text{Initial Equity Investment}} \end{aligned}$$

Allows the analyst to project ROI across an entire range of hypothetical prices without recomputing each input from scratch.

30. Long Margin Ratio at an Assumed Stock Price:

$$\begin{aligned} &\text{Long Margin Ratio at an Assumed Stock Price} = \\ &\frac{(\text{Shares Purchased} \times \text{Assumed Stock Price}) - \text{Amount Borrowed}}{\text{Shares Purchased} \times \text{Assumed Stock Price}} \end{aligned}$$

Extends the margin-ratio concept across a price range, revealing exactly where the position crosses into margin-call territory.

31. Short-Sale ROI at an Assumed Stock Price:

$$\begin{aligned} &\text{Short-Sale ROI at an Assumed Stock Price} = \\ &\frac{\text{Shares Sold Short} \times (\text{Initial Stock Price} - \text{Assumed Stock Price}) - \text{Shares Sold Short} \times \text{Dividend per Share}}{\text{Initial Margin Deposit}} \end{aligned}$$

Projects short-sale returns across alternative price scenarios, essential for stress-testing a bearish position.

32. Short-Sale Margin Ratio at an Assumed Stock Price:

$$\begin{aligned} &\text{Short-Sale Margin Ratio at an Assumed Stock Price} = \\ &\frac{\text{Initial Margin Deposit} + \text{Short-Sale Proceeds} - \text{Dividends Paid} - (\text{Shares Sold Short} \times \text{Assumed Stock Price})}{\text{Shares Sold Short} \times \text{Assumed Stock Price}} \end{aligned}$$

Completes the sensitivity toolkit by tracing short-sale collateral adequacy across a spectrum of adverse and favorable price outcomes.

Part II: Two Comprehensive Problems

The two problems below use the exact numerical inputs specified for this workbook, applied respectively to a long position purchased on margin and to a short sale subject to a margin call. Each problem is followed immediately by a complete solution.

Comprehensive Problem 1: Long Position Purchased on Margin

Consider an investor who commits \$10,000 of personal equity capital and borrows an additional \$10,000 from a broker in order to purchase shares of a target company priced at \$50 per share at the outset of the holding period. The position is held for twelve months, during which the broker charges an annual interest rate of 8% on the outstanding margin loan and the company distributes a cash dividend of \$0.60 per share. By the end of the holding period, the stock price fell to \$40 per share. The broker applies a maintenance margin requirement of 30% on long positions financed with borrowed funds.

Required

1. Total stock purchased.
2. Number of shares purchased.
3. Capital gain or loss.
4. Dividends received.
5. Interest on the margin loan.
6. Net income.
7. ROI with margin.
8. Ending market value.
9. Equity in the margin account.
10. Current margin ratio.
11. Margin-call price.
12. Leverage multiple.
13. Margin-call status.
14. Number of shares that could have been purchased without borrowing.
15. ROI without margin.
16. Difference between leveraged and unleveraged ROI.
17. ROI and margin ratio at selected alternative stock prices.
18. Investment and risk interpretation of the results.

Solution

1. Total Stock Purchased: Initial Equity Investment + Amount Borrowed = \$10,000 + \$10,000 = \$20,000

Interpretation: The investor controls a \$20,000 position, twice the size that personal capital alone would permit.

2. Shares Purchased: Total Stock Purchased ÷ Initial Stock Price = \$20,000 ÷ \$50 = 400 shares

Interpretation: Every subsequent dollar figure in this solution is driven by this 400-share position size.

3. Capital Gain or Loss: Shares Purchased × (Ending Price – Initial Price) = 400 × (\$40 – \$50) = –\$4,000

Interpretation: The 20% price decline produces a \$4,000 loss on the underlying shares before financing costs or dividends are considered.

4. Dividends Received: Shares Purchased $\times$ Cash Dividend per Share = $400 \times \$0.60 = \240

Interpretation: Dividend income offers only a modest offset against the capital loss and financing cost.

5. Interest on the Margin Loan: Amount Borrowed $\times$ Annual Rate $\times$ (Months $\div$ 12) = $\$10,000 \times 0.08 \times (12 \div 12) = \800

Interpretation: Because the position is held for the full year, the entire 8% annual rate applies to the \$10,000 loan.

6. Net Income: Capital Gain or Loss + Dividends Received – Interest = $-\$4,000 + \$240 - \$800 = -\$4,560$

Interpretation: All three cash-flow components are unfavorable or cost-generating, so losses compound rather than offset.

7. ROI with Margin: Net Income $\div$ Initial Equity Investment = $-\$4,560 \div \$10,000 = -0.456$, or -45.6%

Interpretation: Leverage has converted a 20% price decline into a 45.6% loss on the investor's own capital — more than double the underlying price move.

8. Ending Market Value: Shares Purchased $\times$ Ending Price = $400 \times \$40 = \$16,000$

Interpretation: This is the value of collateral available to support the \$10,000 broker loan at period end.

9. Equity in the Margin Account: Ending Market Value – Amount Borrowed = $\$16,000 - \$10,000 = \$6,000$

Interpretation: The investor's residual claim has fallen from an initial \$10,000 to \$6,000, a direct consequence of the price decline.

10. Current Margin Ratio: (Ending Market Value – Amount Borrowed) $\div$ Ending Market Value = $\$6,000 \div \$16,000 = 0.375$, or 37.5%

Interpretation: The margin ratio has fallen from an initial 50% to 37.5%, moving closer to — but not yet breaching — the 30% maintenance floor.

11. Margin-Call Price: Amount Borrowed $\div$ [Shares $\times$ (1 – Maintenance Margin)] = $\$10,000 \div [400 \times 0.70] = \$10,000 \div 280 = \$35.71$

Interpretation: If the stock falls to \$35.71 or below, the account's equity cushion will have eroded to exactly the maintenance threshold.

12. Leverage Multiple: Total Stock Purchased $\div$ Initial Equity Investment = $\$20,000 \div \$10,000 = 2.0\times$

Interpretation: Each dollar of the investor's capital controls two dollars of market exposure, which is precisely why the 20% price decline produced a 45.6% loss on capital.

13. Margin-Call Status: Ending Price \$40 compared with Margin-Call Price \$35.71: since $\$40 > \35.71 , Status = Safe

Interpretation: Despite the substantial loss, the position has not yet triggered a margin call, though the cushion above the critical price has narrowed considerably.

14. Shares Purchased without Margin: Initial Equity Investment $\div$ Initial Stock Price = $\$10,000 \div \$50 = 200$ shares

Interpretation: This establishes the unleveraged benchmark position against which the margin trade is compared.

15. ROI without Margin: $[200 \times (\$40 - \$50) + 200 \times \$0.60] \div \$10,000 = (-\$2,000 + \$120) \div \$10,000 = -0.188$, or -18.8%

Interpretation: Without borrowing, the same adverse price move would have cost the investor only 18.8% of capital, versus 45.6% with leverage employed.

16. Difference between Leveraged and Unleveraged ROI: $-45.6\% - (-18.8\%) = -26.8$ percentage points

Interpretation: Leverage has amplified the loss on capital by 26.8 percentage points — the cost of the additional exposure once financing charges are included.

17. ROI and Margin Ratio at Selected Alternative Stock Prices: See the sensitivity table below, computed for ending prices of \$20 to \$80 in increments of \$5, holding all other inputs constant.

Interpretation: The table isolates the price at which the position moves from loss to profit and shows how the margin ratio recovers as the price rises.

18. Investment and Risk Interpretation

Interpretation: This position illustrates the double-edged nature of margin trading: the 2.0× leverage multiple that would have doubled a gain to roughly 40% under a symmetrically favorable move instead more than doubled the loss on capital once financing costs are layered on top of a 20% price decline. The margin ratio at \$40 (37.5%) remains above the 30% maintenance level, so no immediate broker action is required, but the sensitivity table shows that a further decline to the mid-\$30s would trigger a margin call. An investor evaluating this position should weigh whether the residual 7.5-percentage-point cushion above the maintenance floor justifies continuing to hold, given that leverage has already converted a moderate price decline into a near-halving of invested capital.

Sensitivity Analysis

Ending Stock Price	ROI with Margin	ROI without Margin	Current Margin Ratio
\$20.00	-125.60%	-58.80%	-25.00%
\$25.00	-105.60%	-48.80%	0.00%
\$30.00	-85.60%	-38.80%	16.67%
\$35.00	-65.60%	-28.80%	28.57%
\$40.00	-45.60%	-18.80%	37.50%
\$45.00	-25.60%	-8.80%	44.44%
\$50.00	-5.60%	1.20%	50.00%
\$55.00	14.40%	11.20%	54.55%
\$60.00	34.40%	21.20%	58.33%
\$65.00	54.40%	31.20%	61.54%
\$70.00	74.40%	41.20%	64.29%
\$75.00	94.40%	51.20%	66.67%
\$80.00	114.40%	61.20%	68.75%

Comprehensive Problem 2: Short Sale and Margin Call

An investor who is bearish on a particular stock instructs her broker to sell short 1,000 shares at an initial price of \$100 per share. The broker imposes an initial margin requirement of 50% of the short-sale proceeds, which the investor satisfies with a \$50,000 margin deposit held in the account. The stock pays no dividends during the period the position is held, and the broker requires a 30% maintenance margin. By the time the investor closes the position, the stock price has fallen to \$70 per share.

Required

1. Short sale proceeds.
2. Current value of shares owed.
3. Capital gain or loss.
4. Dividends payable.
5. Net income.
6. ROI on the margin deposit.
7. Effective initial margin percentage.
8. Equity in the short-sale account.
9. Current short-sale margin ratio.
10. Margin-call price.
11. Margin-call status.
12. ROI and margin ratio at selected alternative stock prices.
13. Risk and decision implications for the short seller.

Solution

1. Short-Sale Proceeds: Initial Stock Price $\times$ Shares Sold Short = $\$100 \times 1,000 = \$100,000$

Interpretation: This is the cash credited to the investor's account the moment the borrowed shares are sold.

2. Current Value of Shares Owed: Ending Stock Price $\times$ Shares Sold Short = $\$70 \times 1,000 = \$70,000$

Interpretation: The repurchase liability has fallen by \$30,000 as the stock price declined — a favorable development for the short-seller.

3. Capital Gain or Loss: Shares Sold Short $\times$ (Initial Price – Ending Price) = $1,000 \times (\$100 - \$70) = \$30,000$

Interpretation: The 30% price decline generates a \$30,000 gain, since the short-seller profits directly from falling prices.

4. Dividends Payable: Shares Sold Short $\times$ Cash Dividend per Share = $1,000 \times \$0 = \0

Interpretation: No dividend obligation arises because the stock pays no dividend during the holding period.

5. Net Income: Capital Gain or Loss – Dividends Paid = $\$30,000 - \$0 = \$30,000$

Interpretation: With no dividend obligation to offset the gain, the full capital gain flows through to net income.

6. ROI on the Margin Deposit: Net Income $\div$ Initial Margin Deposit = $\$30,000 \div \$50,000 = 0.60$, or 60%

Interpretation: The 30% price decline has been converted into a 60% return on the collateral actually posted — a leverage multiple of exactly 2 $\times$.

7. Effective Initial Margin Percentage: Initial Margin Deposit $\div$ Short-Sale Proceeds = $\$50,000 \div \$100,000 = 0.50$, or 50%

Interpretation: The investor's deposit exactly matches the broker's stated 50% initial margin requirement, confirming the position was properly collateralized at inception.

8. Equity in the Short-Sale Account: Deposit + Proceeds – Dividends Paid – Value of Shares Owed = $\$50,000 + \$100,000 - \$0 - \$70,000 = \$80,000$

Interpretation: Account equity has grown from an initial \$50,000 to \$80,000 as the price decline reduced the repurchase liability.

9. Current Short-Sale Margin Ratio: Short-Sale Account Equity $\div$ Current Value of Shares Owed = $\$80,000 \div \$70,000 = 1.1429$, or 114.29%

Interpretation: The margin ratio has risen well above its initial 50% level, indicating an increasingly comfortable collateral position as the trade moves in the short-seller's favor.

10. Margin-Call Price: (Deposit + Proceeds – Dividends Paid) $\div$ [Shares $\times$ (1 + Maintenance Margin)] = $\$150,000 \div [1,000 \times 1.30] = \$150,000 \div 1,300 = \$115.38$

Interpretation: If the stock rises to \$115.38 or above, the margin ratio will fall to exactly the 30% maintenance threshold, triggering a margin call.

11. Margin-Call Status: Ending Price \$70 compared with Margin-Call Price \$115.38: since $\$70 < \115.38 , Status = Safe

Interpretation: The position is far from a margin call, since the price has moved in the direction favorable to the short-seller rather than against it.

12. ROI and Margin Ratio at Selected Alternative Stock Prices: See the sensitivity table below, computed for ending prices of \$10 to \$170 in increments of \$10, holding all other inputs constant.

Interpretation: The table traces the full range from the maximum favorable outcome (price near zero) to the point at which losses erode all posted margin.

13. Risk and Decision Implications for the Short-Seller

Interpretation: This trade illustrates the asymmetric risk profile inherent to short selling: the maximum conceivable gain is capped near 200% of the deposit (if the price fell to zero), whereas losses have no theoretical ceiling as price rises without bound, as the sensitivity table below confirms by turning increasingly negative beyond \$150. At the current \$70 price, the position remains highly profitable and safely collateralized, but the substantial distance to the \$115.38 margin-call price should not be mistaken for permanent safety — a sharp reversal driven by short covering or unexpected news could close that gap quickly. A disciplined short-seller would consider a stop-buy order or partial covering to lock in gains rather than assuming the current favorable margin ratio will persist indefinitely.

Sensitivity Analysis

Ending Stock Price	ROI on Margin Deposit	Current Margin Ratio
\$10.00	180.00%	1,400.00%
\$20.00	160.00%	650.00%
\$30.00	140.00%	400.00%
\$40.00	120.00%	275.00%
\$50.00	100.00%	200.00%

Ending Stock Price	ROI on Margin Deposit	Current Margin Ratio
\$60.00	80.00%	150.00%
\$70.00	60.00%	114.29%
\$80.00	40.00%	87.50%
\$90.00	20.00%	66.67%
\$100.00	0.00%	50.00%
\$110.00	-20.00%	36.36%
\$120.00	-40.00%	25.00%
\$130.00	-60.00%	15.38%
\$140.00	-80.00%	7.14%
\$150.00	-100.00%	0.00%
\$160.00	-120.00%	-6.25%
\$170.00	-140.00%	-11.76%

Part III: Short Application Problems

Eighteen short, self-contained problems follow, three under each of the six calculation groups introduced in Part I. Each problem is preceded by a distinct learning objective and followed immediately by a concise, fully worked solution.

Group 1: Position Construction and Financial Exposure

Problem 1.1

Learning Objective: Estimating the effective market exposure created by borrowing funds for a long-margin position.

Problem: An investment manager commits \$18,000 of personal capital and borrows \$27,000 from a prime broker to purchase Continental Materials Corp shares at \$75. Determine the total stock purchased, the number of shares acquired, and the multiple leverage this financing structure creates for the manager's capital.

Solution

Total Stock Purchased: Initial Equity Investment + Amount Borrowed = \$18,000 + \$27,000 = \$45,000

Shares Purchased: Total Stock Purchased ÷ Initial Stock Price = \$45,000 ÷ \$75 = 600 shares

Leverage Multiple: Total Stock Purchased ÷ Initial Equity Investment = \$45,000 ÷ \$18,000 = 2.5×

Interpretation: Every \$1 of the manager's capital controls \$2.50 of stock exposure; a 2.5× leverage multiple means price movements will be amplified two-and-a-half-fold when measured against the manager's own equity base.

Problem 1.2

Learning Objective: Comparing short-sale proceeds against the current value of shares owed as prices move over time.

Problem: A hedge fund sells short 2,400 shares of Meridian Utilities plc at \$42 per share. Eight weeks later, the shares trade at \$37. Calculate the original short-sale proceeds and the current value of shares owed, and state whether the fund's notional exposure has expanded or contracted.

Solution

Short-Sale Proceeds: Initial Stock Price × Shares Sold Short = \$42 × 2,400 = \$100,800

Current Value of Shares Owed: Ending Stock Price × Shares Sold Short = \$37 × 2,400 = \$88,800

Interpretation: The value of shares owed has contracted by \$12,000 relative to the original proceeds, meaning the fund's repurchase obligation has shrunk — a favorable development for a short position whose profit depends on price decline.

Problem 1.3

Learning Objective: Assessing whether an investor's actual margin deposit satisfies the broker's stated initial margin requirement on a short sale.

Problem: An investor sells short 800 shares of Apex Robotics Inc. at \$60 and deposits \$28,000 as margin with the broker. The broker's stated initial margin requirement on short sales is 55% of proceeds. Determine the effective initial margin percentage and assess whether the deposit satisfies the requirement.

Solution

Short-Sale Proceeds: \$60 × 800 = \$48,000

Effective Initial Margin Percentage: Initial Margin Deposit ÷ Short-Sale Proceeds = \$28,000 ÷ \$48,000 = 0.5833, or 58.33%

***Interpretation:** At 58.33%, the investor's deposit exceeds the broker's 55% requirement, so the short position is adequately margined at initiation, and no additional funding is needed to open the trade.*

Group 2: Income, Costs, and Profit Measurement

Problem 2.1

***Learning Objective:** Demonstrating that a favorable stock-price movement does not necessarily produce an attractive net outcome once financing costs are included.*

Problem: An investor buys 500 shares of Silverline Pharmaceuticals at \$80 using \$20,000 own capital and \$20,000 borrowed at 12% for twelve months. The stock pays no dividend and rises to \$83 by year-end. Compute the capital gain, interest cost, and net income.

Solution

Capital Gain: $500 \times (\$83 - \$80) = \$1,500$

Interest on Margin Loan: $\$20,000 \times 0.12 \times (12 \div 12) = \$2,400$

Net Income: $\$1,500 + \$0 - \$2,400 = -\900

***Interpretation:** Despite a positive 3.75% price move, financing costs exceed capital gain, producing a net loss — clear evidence that a favorable price movement alone does not guarantee an attractive outcome once leverage costs are included.*

Problem 2.2

***Learning Objective:** Quantifying the combined effect of dividend income and a short holding period on net income from a long-margin position.*

Problem: An investor purchases 1,000 shares of BlueWave Renewables at \$30 using \$15,000 of their own funds and \$15,000 borrowed at 7% annual interest for a four-month holding period. The stock pays \$0.40 per-share dividend and ends the period at \$33. Compute net income.

Solution

Capital Gain: $1,000 \times (\$33 - \$30) = \$3,000$

Dividends Received: $1,000 \times \$0.40 = \400

Interest on Margin Loan: $\$15,000 \times 0.07 \times (4 \div 12) = \350

Net Income: $\$3,000 + \$400 - \$350 = \$3,050$

***Interpretation:** Dividend income partially offsets financing costs, and because the holding period is short, the interest drag is modest, allowing most of the capital gain to convert into net income.*

Problem 2.3

***Learning Objective:** Evaluating how a dividend obligation affects net income on a profitable short position.*

Problem: A trader sells short 600 shares of Deccan Energy Ltd at \$55, when the broker requires an initial margin of 50%. During the short position, the company pays a \$0.75 per-share dividend, and the trader later covers \$47. Compute capital gain, dividend obligation, and net income.

Solution

Capital Gain: $\text{Shares Sold Short} \times (\text{Initial Price} - \text{Ending Price}) = 600 \times (\$55 - \$47) = \$4,800$

Dividends Paid: $600 \times \$0.75 = \450

Net Income: $\$4,800 - \$450 = \$4,350$

Interpretation: Even though the trader must compensate the lender for the dividend, the price decline is large enough that the short position remains highly profitable; dividend obligations matter more when the anticipated price decline is small.

Group 3: Return and Leverage Analysis

Problem 3.1

Learning Objective: Comparing ROI with margin against ROI without margin to quantify the return-amplifying effect of leverage.

Problem: An investor with \$25,000 can buy 500 shares of Northfield Industrials outright at \$50, or use the same capital as margin, borrowing \$25,000 at 10% for one year to buy 1,000 shares. If the stock rises to \$58 with no dividend, compute ROI under both approaches.

Solution

ROI without Margin: $[500 \times (\$58 - \$50)] \div \$25,000 = \$4,000 \div \$25,000 = 0.16$, or 16%

Interest on Margin Loan: $\$25,000 \times 0.10 \times 1 = \$2,500$

Net Income (margin case): $1,000 \times (\$58 - \$50) - \$2,500 = \$8,000 - \$2,500 = \$5,500$

ROI with Margin: $\$5,500 \div \$25,000 = 0.22$, or 22%

Interpretation: Leverage lifts ROI from 16% to 22% on identical capital because the doubled share position outweighs the financing cost — the return-amplification effect of borrowing at a rate below the stock's rate of appreciation.

Problem 3.2

Learning Objective: Reconciling total accounting profit with the return earned specifically on the investor's own committed capital, to guide a hold decision.

Problem: A portfolio manager buys 2,000 shares of Continental Materials Corp at \$45 using \$45,000 own capital and \$45,000 borrowed at 8% for six months. The position generates a \$9,000 net income when closed. The manager must decide whether this outcome justifies the leverage taken. Compute ROI and advise.

Solution

ROI with Margin: $\text{Net Income} \div \text{Initial Equity Investment} = \$9,000 \div \$45,000 = 0.20$, or 20%

Interpretation: The \$9,000 figure looks like a modest 10% profit relative to the \$90,000 position size, but it represents a 20% return on the manager's own \$45,000 once leverage is properly credited to the smaller equity base. The decision should rest on this 20% capital-adjusted return, not the smaller headline profit-to-position ratio, and the outcome appears to justify the risk taken.

Problem 3.3

Learning Objective: Evaluating short-sale ROI relative to margin deposit and comparing it with the underlying percentage price decline to reveal embedded leverage.

Problem: A trader deposits \$40,000 margin to sell short 1,600 shares of Vertex Pharmaceuticals at \$50, under a 50% initial margin requirement. The stock falls to \$42 with no dividend obligation. Compute the short-sale ROI and compare it with the percentage price decline to illustrate the position's leverage.

Solution

Capital Gain / Net Income: $1,600 \times (\$50 - \$42) = \$12,800$

Short-Sale ROI: $\$12,800 \div \$40,000 = 0.32$, or 32%

Interpretation: The stock declined only 16%, yet the short position returned 32% on the margin deposit — a leverage multiple of exactly 2× — confirming that a 50% margin requirement doubles the percentage return relative to the underlying price movement.

Group 4: Account Equity and Margin Adequacy

Problem 4.1

Learning Objective: Tracking account equity and margin-ratio deterioration on a long position that remains above the maintenance threshold.

Problem: An investor holds 900 shares of Northfield Industrials, purchased at \$60, using \$27,000 of own capital and \$27,000 borrowed. The maintenance margin is 25%. The price falls to \$52. Compute the ending market value, account equity, and current margin ratio, and confirm whether the position remains adequately margined.

Solution

Ending Market Value: $900 \times \$52 = \$46,800$

Account Equity: $\$46,800 - \$27,000 = \$19,800$

Current Margin Ratio: $\$19,800 \div \$46,800 = 0.423$, or 42.3%

Interpretation: Although the margin ratio has fallen from an initial 50% to 42.3%, it remains comfortably above the 25% maintenance threshold, so no margin call is triggered despite the price decline.

Problem 4.2

Learning Objective: Identify a scenario in which account equity turns negative and collateral becomes critically inadequate.

Problem: An investor purchased 1,000 shares of Meridian Utilities plc at \$40, contributing \$20,000 and borrowing \$20,000. Following an adverse earnings announcement, the stock collapses to \$18. Compute the ending market value and equity value, and explain the implications of a negative equity outcome for the broker and investor.

Solution

Ending Market Value: $1,000 \times \$18 = \$18,000$

Account Equity: $\$18,000 - \$20,000 = -\$2,000$

Interpretation: Negative equity means the stock's value no longer covers the broker's loan; the investor is effectively insolvent on this position, and the collateral has already been exhausted well before this point — a scenario reflecting a serious control failure, since a margin call and forced liquidation should have occurred long before equity turned negative.

Problem 4.3

Learning Objective: Evaluating short-sale account equity and margin-ratio adequacy following an adverse price increase.

Problem: A short-seller deposits \$35,000 margin and sells short 700 shares of Apex Robotics Inc. at \$50, receiving \$35,000 in proceeds. No dividend is paid, but the stock later rises to \$58. Compute the current value of shares owned, short-sale account equity, and the current short-sale margin ratio.

Solution

Current Value of Shares Owed: $700 \times \$58 = \$40,600$

Short-Sale Account Equity: $\$35,000 + \$35,000 - \$0 - \$40,600 = \$29,400$

Current Short-Sale Margin Ratio: $\$29,400 \div \$40,600 = 0.724$, or 72.4%

Interpretation: Despite the 16% adverse price rise, the margin ratio remains high because the initial margin cushion was generous; the position stays well collateralized and far from a margin call.

Group 5: Margin-Call Thresholds and Risk Control

Problem 5.1

Learning Objective: Deriving the critical margin-call price for a long-margin position.

Problem: An investor buys 750 shares of BlueWave Renewables at \$64 using \$24,000 own capital and \$24,000 borrowed from the broker. The maintenance margin requirement is 35%. Determine the exact stock price at which the position will trigger a margin call.

Solution

Long Margin-Call Price: $\text{Amount Borrowed} \div [\text{Shares} \times (1 - \text{Maintenance Margin})] = \$24,000 \div [750 \times 0.65] = \$24,000 \div 487.5 = \$49.23$

Interpretation: If BlueWave Renewables falls to \$49.23 or below, the account's equity cushion will have eroded to exactly the 35% maintenance threshold, triggering a margin call.

Problem 5.2

Learning Objective: Comparing the effect of two different maintenance-margin policies on the short-sale margin-call price, to assess the risk implication of a broker tightening its requirement.

Problem: A short-seller deposits \$45,000 to sell short 900 shares of Deccan Energy Ltd at \$50, generating \$45,000 proceeds, with no dividend. Determine the margin-call price under a 30% maintenance margin and again if the broker raises the requirement to 40% and compares the risk implications.

Solution

Margin-Call Price at 30% Maintenance: $(\$45,000 + \$45,000 - \$0) \div [900 \times 1.30] = \$90,000 \div 1,170 = \$76.92$

Margin-Call Price at 40% Maintenance: $\$90,000 \div [900 \times 1.40] = \$90,000 \div 1,260 = \$71.43$

Interpretation: Raising the maintenance margin from 30% to 40% lowers the trigger price from \$76.92 to \$71.43, tightening the price cushion the short-seller has before facing a margin call — a stricter policy gives the broker earlier warning but leaves the trader less room for adverse price moves.

Problem 5.3

Learning Objective: Determining margin-call status at a given ending price and identifying the appropriate risk-control response.

Problem: An investor holds a long-margin position of 1,200 shares of Continental Materials Corp bought at \$35, borrowing \$18,000 against \$24,000 own capital, with a 30% maintenance margin. The price has fallen to \$28. Determine the margin call price, assess the current status, and recommend an appropriate action.

Solution

Long Margin-Call Price: $\$18,000 \div [1,200 \times 0.70] = \$18,000 \div 840 = \$21.43$

Status: Ending Price \$28 > Margin-Call Price \$21.43, so Status = Safe

Interpretation: Although the position has lost value, the margin ratio remains above the maintenance threshold; the current status is safe, and no immediate action is required, though the investor should monitor closely since the cushion above the call price has narrowed considerably.

Group 6: Sensitivity and Decision Analysis

Problem 6.1

Learning Objective: Evaluating how ROI on a long-margin position changes across a narrow band of alternative stock prices to identify the break-even region.

Problem: An investor purchases 500 shares of Vertex Pharmaceuticals at \$40 using \$10,000 of own capital and \$10,000 borrowed at 8% for 1 year, with no dividends. Compute the ROI if the stock ends at \$36, \$40, and \$44, and identify the approximate break-even price.

Solution

Interest on Margin Loan: $\$10,000 \times 0.08 \times 1 = \800

ROI at \$36: $[500 \times (\$36 - \$40) - \$800] \div \$10,000 = (-\$2,000 - \$800) \div \$10,000 = -28\%$

ROI at \$40: $[\$0 - \$800] \div \$10,000 = -8\%$

ROI at \$44: $[500 \times \$4 - \$800] \div \$10,000 = (\$2,000 - \$800) \div \$10,000 = 12\%$

Interpretation: Break events occur where capital gain equals financing cost, i.e., $500 \times (P - \$40) = \800 , giving $P = \$41.60$.

Below this price, the position loses money even though the stock itself may be little changed, illustrating how financing cost shifts the effective break-even point above the purchase price.

Problem 6.2

Learning Objective: Comparing the long margin ratio across alternative prices to determine when a position approaches its maintenance threshold.

Problem: An investor holds 600 shares of Silverline Pharmaceuticals, bought at \$50, and borrows \$15,000 against \$15,000 of own capital, with a 30% maintenance margin. Compute the margin ratio if the price falls to \$38, \$34, and \$30, and state at which of these prices a margin call occurs.

Solution

Margin Ratio at \$38: $(600 \times \$38 - \$15,000) \div (600 \times \$38) = \$7,800 \div \$22,800 = 34.2\%$

Margin Ratio at \$34: $(600 \times \$34 - \$15,000) \div (600 \times \$34) = \$5,400 \div \$20,400 = 26.5\%$

Margin Ratio at \$30: $(600 \times \$30 - \$15,000) \div (600 \times \$30) = \$3,000 \div \$18,000 = 16.7\%$

Interpretation: The margin ratio crosses the 30% maintenance threshold at \$35.71 (the exact margin-call price); consequently, the position remains safe at \$38 but is already subject to a margin call at both \$34 and \$30.

Problem 6.3

Learning Objective: Evaluating short-sale ROI and margin ratio jointly across alternative prices to recommend whether to hold or cover the position.

Problem: A trader sells short 500 shares of Northfield Industrials at \$80 with a \$25,000 margin deposit and a 30% maintenance margin. No dividend is expected. Compute the ROI and margin ratio for prices of \$70, \$80, and \$95, and recommend whether to hold or cover at each level.

Solution

At \$70: Net Income = $500 \times (\$80 - \$70) = \$5,000$; ROI = $\$5,000 \div \$25,000 = 20\%$; Margin Ratio = $(\$25,000 + \$40,000 - \$35,000) \div \$35,000 = 85.7\%$

At \$80: Net Income = \$0; ROI = 0%; Margin Ratio = $(\$25,000 + \$40,000 - \$40,000) \div \$40,000 = 62.5\%$

At \$95: Net Income = $500 \times (\$80 - \$95) = -\$7,500$; ROI = $-\$7,500 \div \$25,000 = -30\%$; Margin Ratio = $(\$25,000 + \$40,000 - \$47,500) \div \$47,500 = 36.8\%$

***Interpretation:** At \$70 and \$80, the position remains profitable or flat with ample margin cushion, supporting a hold decision. At \$95, the position is down 30% on the deposit, and the margin ratio has fallen to 36.8%, just above the 30% maintenance floor — since the margin-call price is \$100, the trader is not yet forced to act, but prudent risk management would favor covering part of the position or posting a stop-buy order before the cushion disappears entirely.*